

B. Tech End Semester Examination Dec 2025
1st Semester
Basic Electronics Engineering

Time: Three Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks for each main question is twenty.
(iv) Each question carries 10 marks.

Q1	(20 marks)	CO1
(a)	(i) Convert the given Hexadecimal number to Binary, Octal and Decimal. $(4CB2)_{16} = (?)_2 = (?)_8 = (?)_{10}$ (ii) Convert the given Decimal number to Binary, Octal and Hexadecimal $(345)_{10} = (?)_2 = (?)_8 = (?)_{16}$	
(b)	(i) State and prove De Morgan theorem. (ii) Draw the logic circuit for the given expression using basic logic gates. $AB + BC' + A'BC$	
(c)	Simplify the following expression using K – Map and relase the output with NAND gate only. $Y(A, B, C, D) = \sum m(0, 6, 8, 13, 14) + \sum d(2, 4, 10)$	
Q2	(20 marks)	CO2
(a)	Define current amplification factors α and β in bipolar junction transistor? In a certain transistor collector current (I_C) is 6mA, base current (I_B), is 60 μ A and I_{CBO} is 1.5 μ A. Determine the values of α and β .	
(b)	(i) Describe working of PN diode under forward and reverse bias condition with its complete V-I characteristics graph. (ii) Determine the diode current at room temperature for Si diode with reverse saturation current 100 nA and applied forward voltage of 0.52 volt. Consider ideality factor η as 2 and thermal voltage (V_T) as 26mV.	
(c)	Determine the conductivity of pure Si at room temperature with intrinsic concentration is $1.6 \times 10^{10} / \text{cm}^3$ and the mobility of electrons and holes in Si are 1500 $\text{cm}^2/\text{V-s}$ and 500 $\text{cm}^2/\text{V-s}$ respectively. Calculate the conductivity of Silicon, if N type donor impurity is added to the extent of 1 impurity atom in 10^8 Si atoms. Density of Si atom is $2.0 \times 10^{23} / \text{cm}^3$.	
Q3	(20 marks)	CO2, CO1
(a)	Draw the circuit diagrams of Bipolar Junction Transistor (BJT) in Common Base (CB) for n-p-n transistor and explain current amplification factor and total current.	
(b)	(i) Explain classification of materials as conductor, semiconductor and insulator with supporting energy band diagrams. (ii) Compare intrinsic and extrinsic semiconductors with an example for each.	
(c)	Determine the base in following equations: (i) $(28)_{16} = (50)_x$ (ii) $(535)_x = (1333)_{10}$	
Q4	(20 marks)	CO3
(a)	Explain half wave rectifier with diagram and input/output waveforms. Determine the efficiency and ripple factor for the half wave rectifier.	
(b)	A diode having internal resistance of R_f is 20 Ω used for half wave rectification. If the applied voltage $v=100\sin\omega t$ and load resistance R_L is 800 Ω . Determine the I_{max} , I_{dc} , I_{rms} , P_{ac} , P_{dc} .	
(c)	Determine the output voltage of an Inverting amplifier with following parameters: Input volage = 5V, Input Resistance (R_i) = 5K Ω and Feedback resistance (R_f) = 50 K Ω . If the configuration is changed as Non-Inverting amplifier, determine the output voltage.	
Q5	(20 marks)	CO1, CO3
(a)	List the characteristics for ideal op-amp and draw the schematic of inverting and non-inverting configuration.	
(b)	Derive the closed loop voltage gain of both configuration of Op-Amp: Inverting and Non- Inverting amplifier.	
(c)	Define min-terms and max-terms considering A, B and C as Boolean variables. Express the given function as sum of min-terms $F = A + B'C$	

